EEE407 Tutorial 4b

Answer the questions based in the information given in the appendix.

1. List the components of an ACMV system.
2. What are the internal heat gains in a building?
3. Calculate the sensible and latent people heat gain in a lecture hall of 300 m^2.
4. A call centre operates 24 hours a day with the staffs working in shifts. Calculate the total people and lighting (12W/m^2) heat gains to be included in the calculation of cooling capacity, given that the floor area of 500 m^2 and diversity factor of 0.6 (applicable to both heat gains).
5. Calculate the total lowest and highest people heat gain of a 3-level shopping mall with shopping area of 5000 m^2 per level, with maximum occupant density of 2.1 person/m^2. Diversity factor is 0.7 on weekends.
6. Calculate the fresh air requirement per person for an air-conditioned room with 4 persons in an area of 100 m^2, in cfm/person and cfm/ft^2. What is the sensible heat gain added to the cooling coil if the outside temperature is 32°C and 16°C is required after the cooling coil?
7. A fan is used to move air through an air conditioning duct. The air flow rate of 10,200 m^3/hr is required in this application. Compare the fan power needed if the fan pressure is 2 kPa. Compare the power for four types of centrifugal fans as shown in Figure T4-1.

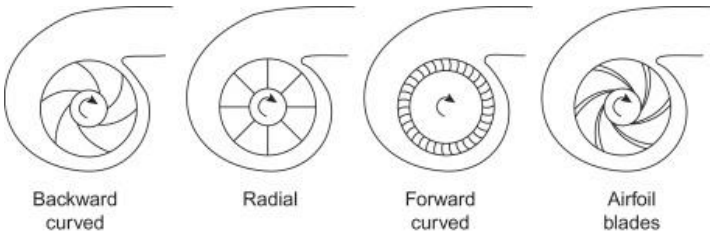


Figure T4-1

8. The temperature sensor and flow gauge of a chiller show the measurement as follow:

   Volume flow rate, qv: 10 m^3/min

   Outlet temperature T1: 6°C

   Return temperature T2: 16°C

   Electrical input power: 750 kW

   Clear water is circulated in the chill water loop. Find the cooling capacity and the coefficient of performance (COP) of this chiller. How is this chiller (rated at 800 kW) performing according to the ASHRAE standard? What are the potential issues and what could be improved?

9. The temperature and humidity of the air is reduced as the air moves through the cooling coil or evaporator of the air conditioner, as shown in Figure T4-2. Given that:

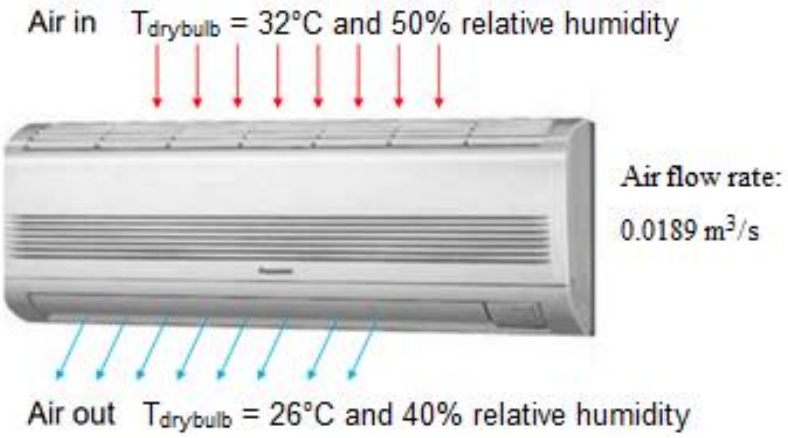


Figure T4-2

a) Use psychrometric chart https://cdn2.hubspot.net/hubfs/331340/Psychrometric-Chart-1.jpg (Appendix Figure 4-3) to determine the humidity ratio in lb (water)/lb (dry air) and enthalpy in btu/lb before and after the cooling coil.

b) Calculate the cooling coil load, $Q_{sensible}$, Q_{latent}, and Q_{total}.

Appendix

Table T4-1: Benchmark allowances for internal heat gains in typical buildings

Building type	Use	Density of occupation / person·m^{-2}	Sensible heat gain / W·m^{-2}			Latent heat gain / W·m^{-2}	
			People	Lighting*	Equip't†	People	Other
Offices	General	12	6.7	8–12	15	5	—
		16	5	8–12	12	4	—
	City centre	6	13.5	8–12	25	10	—
		10	8	8–12	18	6	—
	Trading/dealing	5	16	12–15	40+	12	—
	Call centre floor	5	16	8–12	60	12	—
	Meeting/conference	3	27	10–20	5	20	—
	IT rack rooms	0	0	8–12	200	0	—
Airports/stations‡	Airport concourse	0.83	75	12	5	4	—
	Check-in	0.83	75	12	5	50	—
	Gate lounge	0.83	75	15	5	50	—
	Customs /immigration	0.83	75	12	5	50	—
	Circulation spaces	10	9	12	5	6	—
Retail	Shopping malls	2–5	16–40	6	0	12–30	—
	Retail stores	5	16	25	5	12	—
	Food court	3	27	10	†	20	§
	Supermarkets	5	16	12	†	12	§
	Department stores:						
	— jewellery	10	8	55	5	6	—
	— fashion	10	8	25	5	6	—
	— lighting	10	8	200	5	6	—
	— china/glass	10	8	32	5	6	—
	— perfumery	10	8	45	5	6	—
	— other	10	8	22	5	6	—
Education	Lecture theatres	1.2	67	12	2	50	—
	Teaching spaces	1.5	53	12	10	40	—
	Seminar rooms	3	27	12	5	20	—
Hospitals	Wards	14	57	9	3	4.3	—
	Treatment rooms	10	8	15	3	6	—
	Operating theatres	5	16	25	60	12	—
Leisure	Hotel reception	4	20	10–20	5	15	—
	Banquet/conference	1.2	67	10–20	3	50	—
	Restaurant/dining	3	27	10–20	5	20	—
	Bars/lounges	3	27	10–20	5	20	—

* The internal heat gain allowance should allow for diversity of use of electric lighting coincident with peak heat gain and maximum temperatures. Lighting should be switched off in perimeter/window areas (up to say 4.5 m) and no allowance account for any dimming or other controls.

† Equipment gains do not allow for large duty local equipment such as heavy-duty photocopiers and vending machines.

‡ The exact density will depend upon airport and airplane capacity, the type of gate configuration (open or closed) and passenger throughput. Absolute passenger numbers if available would be a more appropriate design basis. Appropriate building scale diversities need to be derived based on airport passenger throughput.

§ Latent gains are likely but there are no benchmark allowances and heat gains need to be calculated from the sources, e.g. for meals, 15 W per meal[6] served, of which 75% is sensible and 25% latent heat; see also Appendix 6.A2

Table T4-2: Total fresh air requirements per person for different occupant densities.

Occupant Density (m^2/person)	Floor Area Requirement l/s per person	People requirement l/s per person	Total l/s per person*	Total cfm per person*	Total l/s per m^2	Total cfm per ft^2
8	2.4	2.5	4.9	10.4	0.61	0.121
10	3.0	2.5	5.5	11.7	0.55	0.108
12	3.6	2.5	6.1	12.9	0.51	0.100
14	4.2	2.5	6.7	14.2	0.48	0.094
16	4.8	2.5	7.3	15.5	0.46	0.090
18	5.4	2.5	7.9	16.7	0.44	0.086
20	6.0	2.5	8.5	18.0	0.43	0.084

Table T4-3: Minimum fan efficiency for different air flow rates and fan types.

Type of Fan	Category of Fan Type	Air Flow Rate (m^3/hr)	Minimum Fan Total Efficiency (%)
Centrifugal	Forward Curved	1,800	52
		3,400	60
		6,800	62
		10,200	67
		13,600	67
		17,000	67
	Backward Curved	1,800	72
		3,400	76
		6,800	77
		10,200	78
		13,600	78
		17,000	80
	Air Foil	1,800	75
		3,400	79
		6,800	81
		10,200	82
		13,600	83
		17,000	83
	Radial Blade	1,800	60
		3,400	60
		6,800	61
		10,200	63
		13,600	65
		17,000	65

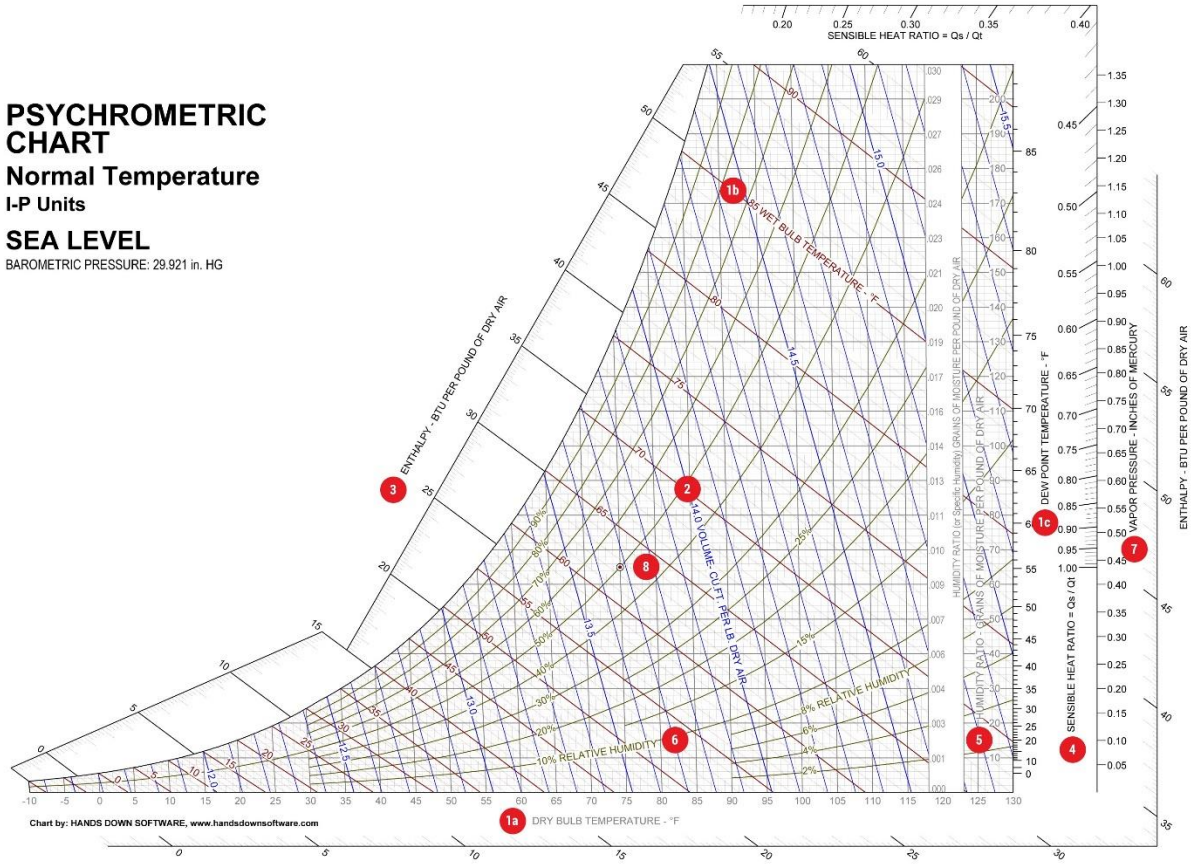

PSYCHROMETRIC CHART
Normal Temperature
I-P Units
SEA LEVEL
BAROMETRIC PRESSURE: 29.921 in. HG
SENSIBLE HEAT RATIO = Qs / Qt
ENTHALPY - BTU PER POUND OF DRY AIR
WET BULB TEMPERATURE - °F
VOLUME CU.FT. PER LB. DRY AIR
RELATIVE HUMIDITY
DEW POINT TEMPERATURE - °F
HUMIDITY RATIO - GRAINS OF MOISTURE PER POUND OF DRY AIR
VAPOR PRESSURE - INCHES OF MERCURY
DRY BULB TEMPERATURE - °F
Chart by: HANDS DOWN SOFTWARE, www.handsdownsoftware.com
1a
1b
1c
2
3
4
5
6
7
8

Figure 4-3